

Student Enrollment Management System Algorithm

1. Start the program.
2. Display the Main Menu.
3. Read the user's menu choice.
4. **If choice is Register Student:**
 - 4.1 Check if a maximum of 50 students has been reached. If yes, stop registration.
 - 4.2 Input Student ID until a unique ID is entered.
 - 4.3 Input Student Name, Course, and Year Level (1-4).
 - 4.4 Initialize Units, Tuition Fee, Laboratory Fee, Miscellaneous Fee, and Total Fee to 0.
 - 4.5 Set Enrollment Status to NOT ENROLLED.
 - 4.6 Save the student record.

5. If choice is Enroll Student:

- 5.1 Input Student ID.
- 5.2 Search for the student using Linear Search.
- 5.3 If not found, display 'Student record not found' and return to menu.
- 5.4 Input Number of Units. If units are below 12 or above 30, cancel enrollment.
- 5.5 Input Laboratory Fee (0-5000). Validate input.
- 5.6 Input Miscellaneous Fee (0-10000). Validate input.
- 5.7 Compute Tuition Fee = Units × 1850.
- 5.8 Compute Total Fees = Tuition + Laboratory + Miscellaneous.
- 5.9 Store values and set status to ENROLLED.

6. If choice is Display All Students:

- 6.1 If no students exist, display a message.
- 6.2 Otherwise, display every student's information.

7. If choice is Search Student:

- 7.1 Input Student ID.
- 7.2 Search using Linear Search.
- 7.3 If found, display complete student information; otherwise display not found.

8. If choice is Enrollment Summary:

- 8.1 Count enrolled and not enrolled students.
- 8.2 Compute total, average, highest, and lowest tuition fees.
- 8.3 Display the summary report.

9. If the choice is Exit, display termination message and end the program.

10. Otherwise, display invalid choice and return to the Main Menu.